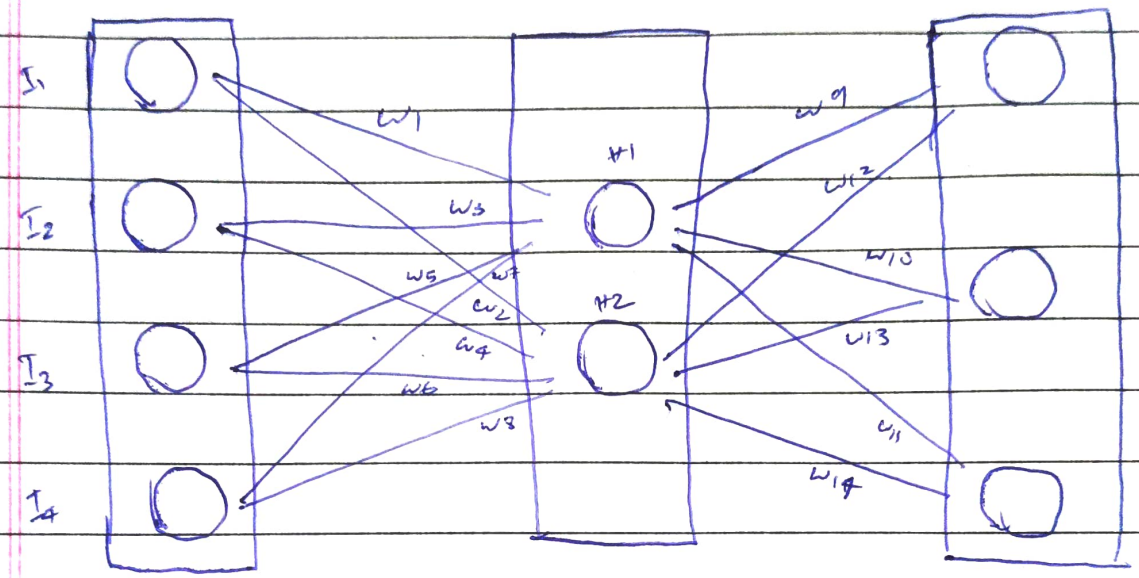
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# Mathematical Calculation
## Single - Hidden - Layer Neural Network




Given Parameters

- Input vector: $X = [5.1, 3.2, 1.4, 0.2]$
- Hidden weights ($H_1$): $w_1 = 1$ $w_3 = 3$
  $w_5 = 5$ $w_7 = 7$
- Hidden weights ($H_2$): $w_2 = 2$ $w_4 = 4$
- $w_6 = 6$ $w_8 = 8$
- Output weights (from $H_1$): $w_9 = 1.3$,
  $w_{10} = 0.5$ $w_{11} = 1.2$
- Output weights (from $H_2$): $w_{12} = 0.5$
  $w_{13} = 0.2$ $w_{14} = 0.4$
- Biases: $b_1 = 2$ $b_2 = 0.1$

2. Hidden Layer Calculation

A) Neuron 1 (H1):

• Linear combination ($z_1$):

$$z_1 = x_1 w_1 + x_2 w_3 + x_3 w_3 + x_4 w_4 + b_1$$

$$z_1 = (5.1 \times 1) + (3.2 \times 3) + (1.4 \times 5) + (0.2 \times 7) + 2$$

$$z_1 = 5.1 + 9.6 + 7 + 1.4 + 2$$

$$z_1 = 25.1$$

• Activation ($\sigma_1$):

$$\sigma_1 = \frac{1}{1 + e^{-z_1}} = \frac{1}{1 + e^{-25.1}}$$

$$\sigma_1 \cong 1.0$$

B) Neuron 2 (H2):

• Linear combination ($z_2$):

$$z_2 = x_1 w_2 + x_2 w_4 + x_3 w_6 + x_4 w_8 + b_1$$

$$z_2 = (5.1 \times 2) + (3.2 \times 4) + (1.6 \times 6) + (0.2 \times 8) \times 2$$

$$z_2 = 35.0$$

• Activation ($\sigma 2$):

$$\sigma_2 = \frac{1}{1+e^{-22}} = \frac{1}{1+e^{-35.0}}$$

$$\sigma_2 \approx 1.0$$

3 Output Layer calculation [combining both neurons]

A) Setosa (Z setosa):

$$Z\ Setosa = (\sigma_1 \times w_9) + (\sigma_2 \times w_{12}) + b_2$$

$$= (1.0 \times 1.3) + (1.0 \times 0.5) + 0.1$$

$$= 1.3 + 0.5 + 0.1$$

$$Z\ Setosa = 1.9$$

B) Versicolor (Z versicolor):

$$Z\ versicolor = (\sigma_1 \times w_{10}) + (\sigma_2 \times w_{13}) + b_2$$

$$= (1.0 \times 0.5) + (1.0 \times 0.2) + 0.1$$

$$= 0.5 + 0.2 + 0.1$$

Z versicolor $= 0.8$

6) Virginica (z virginica)

z virginica = $(\sigma_1 \times w_{11}) + (\sigma_2 \times w_{12}) + b_2$

$= (1.0 \times 1.2) + (1.0 \times 0.4) + 0.1$

$= 1.2 + 0.4 + 0.1$

z virginica = 1.7

4. Final Prediction

- output vector : [1.9, 0.8, 1.7]
- Prediction : Setosa [Highest Score 1.9]